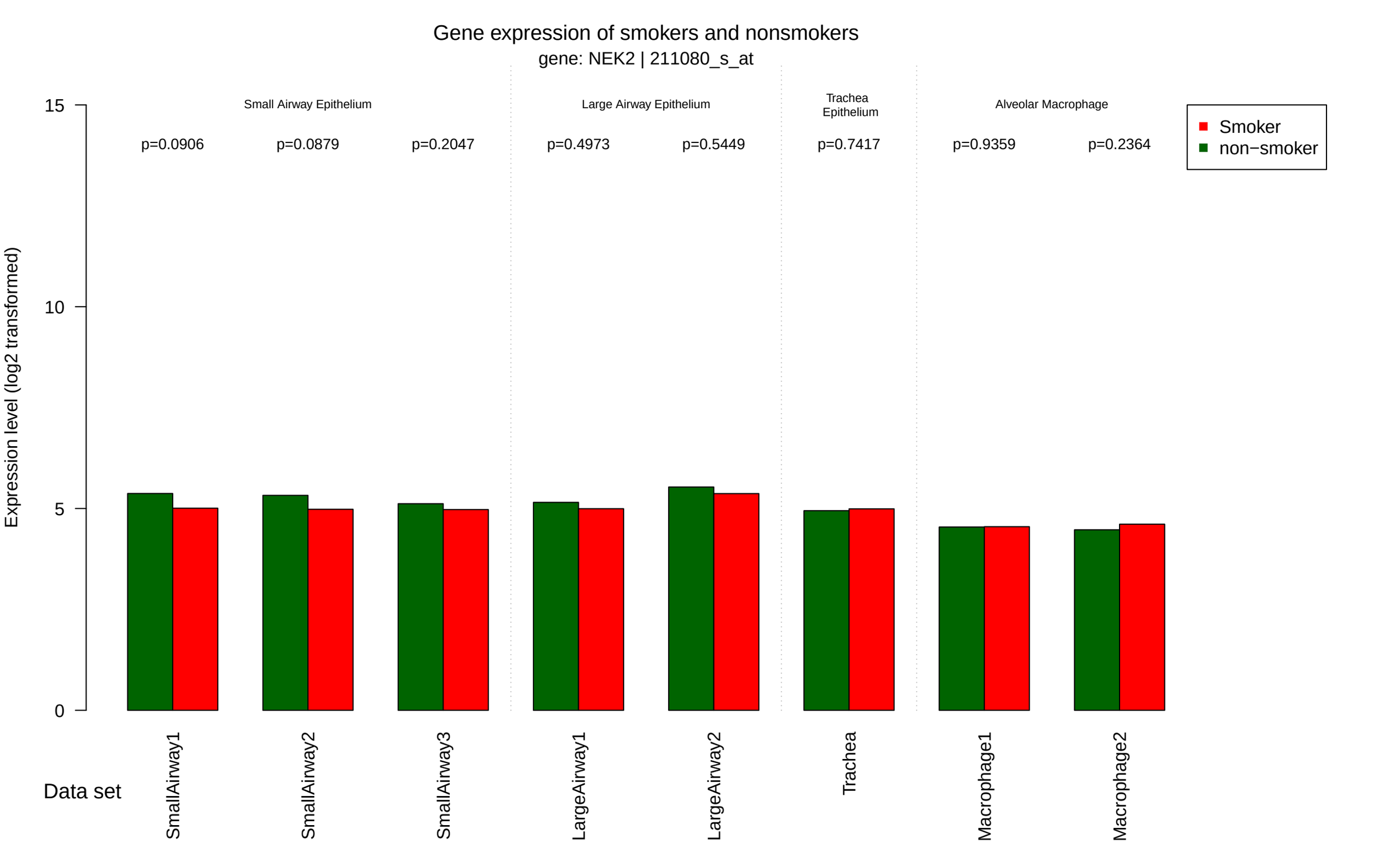
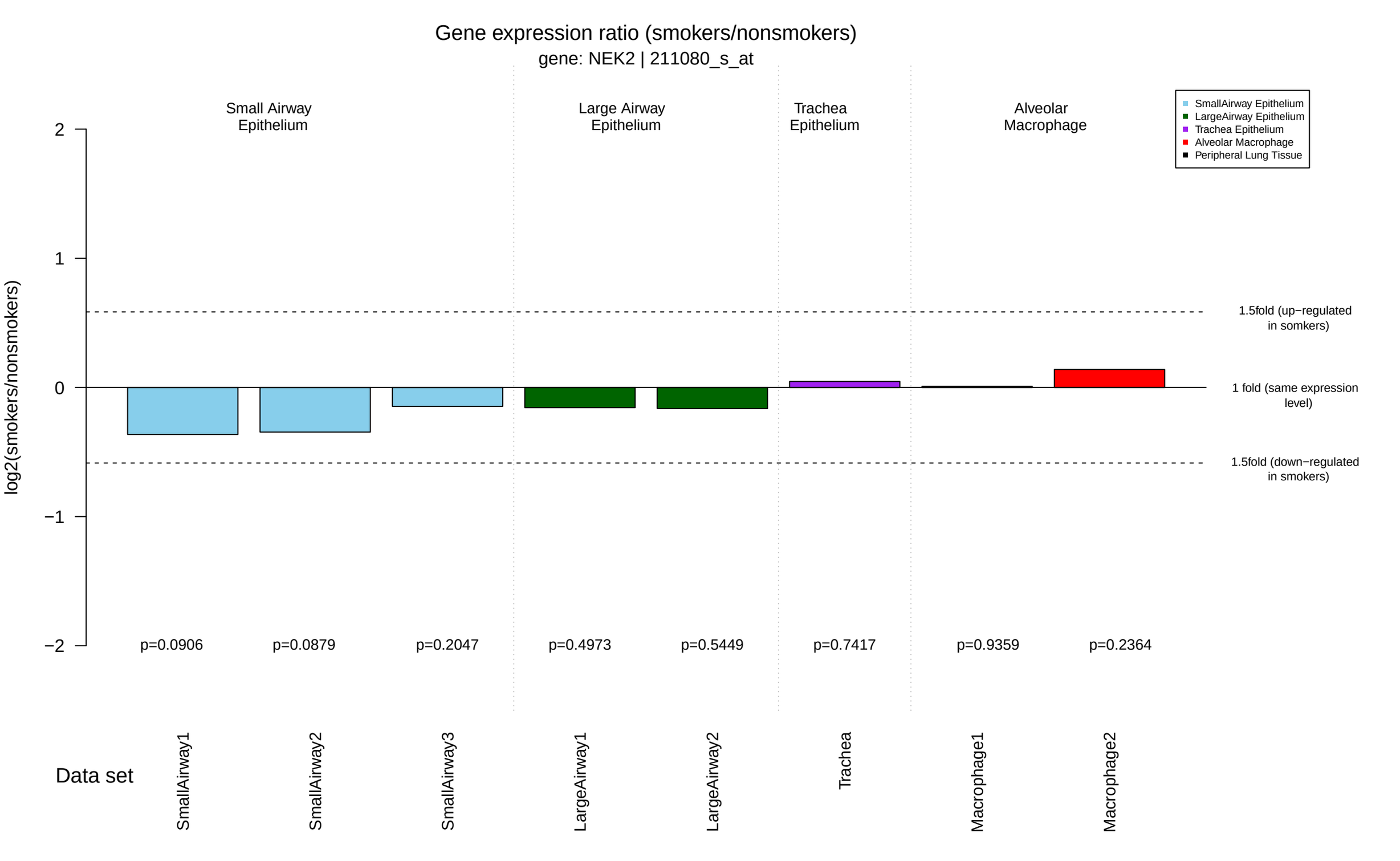
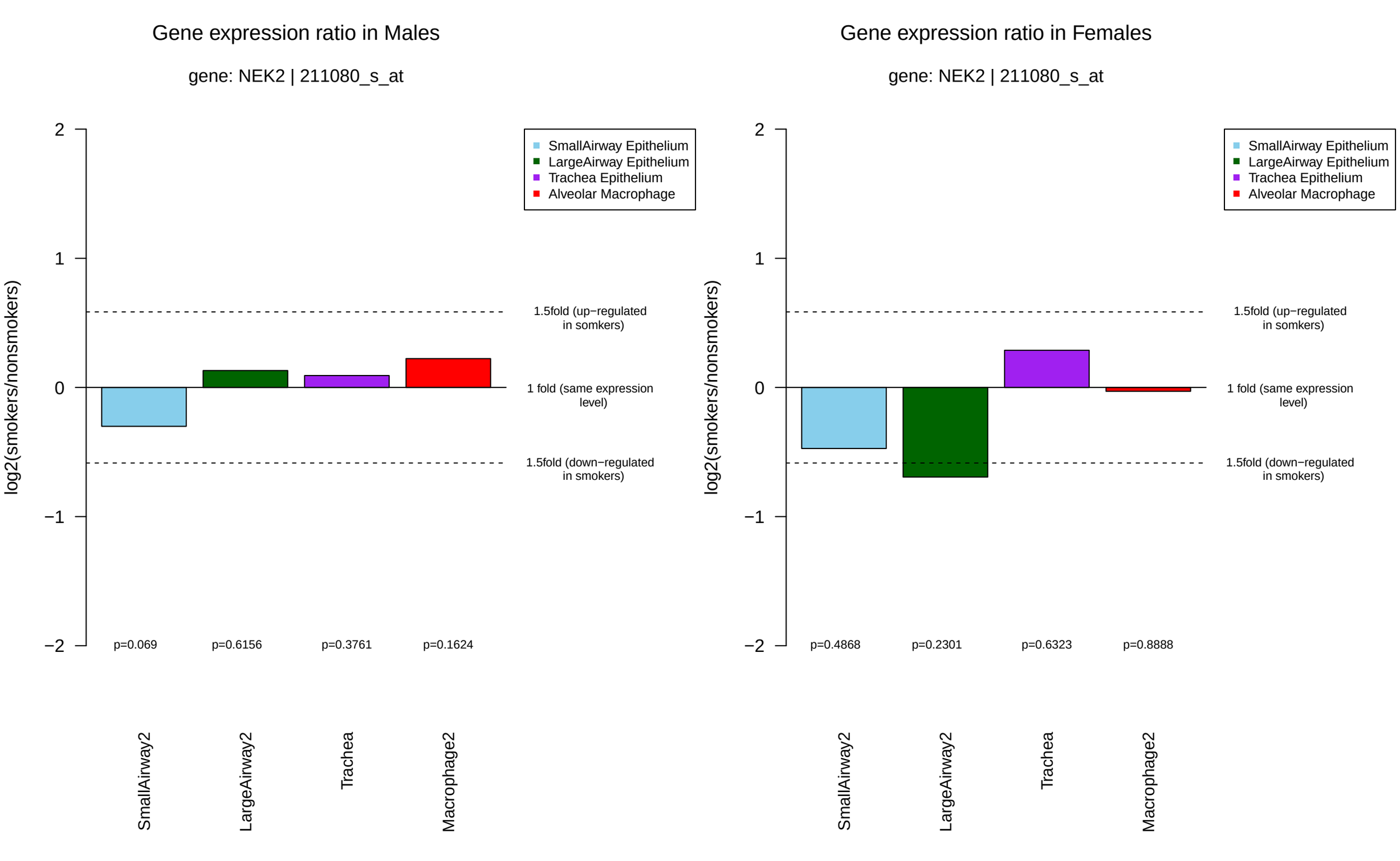
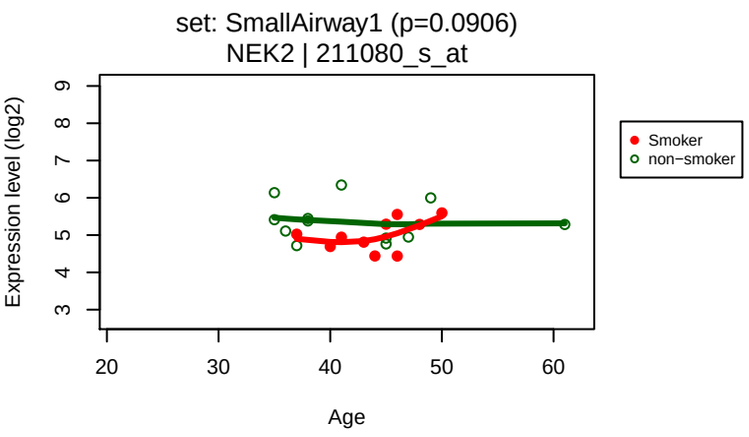
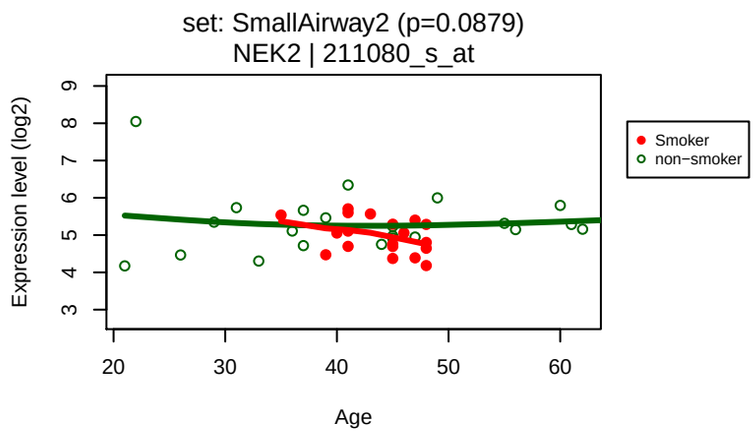
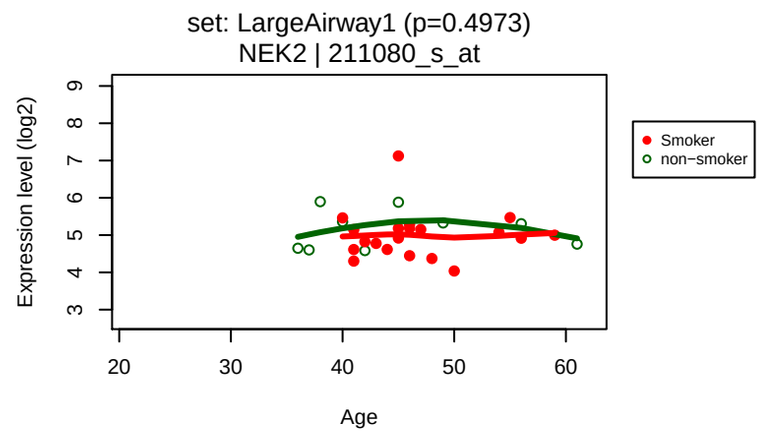

Gene expression of smokers and nonsmokers
gene: NEK2 | 211080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0906
p=0.0879
p=0.2047
p=0.4973
p=0.5449
p=0.7417
p=0.9359
p=0.2364
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

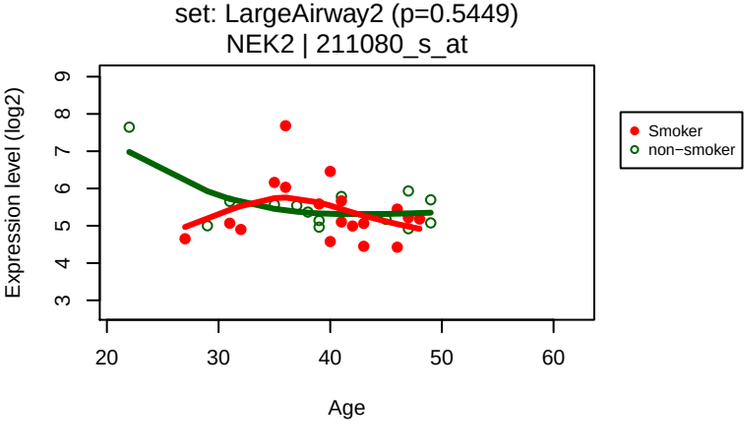

Gene expression ratio (smokers/nonsmokers)
gene: NEK2 | 211080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0906
p=0.0879
p=0.2047
p=0.4973
p=0.5449
p=0.7417
p=0.9359
p=0.2364
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

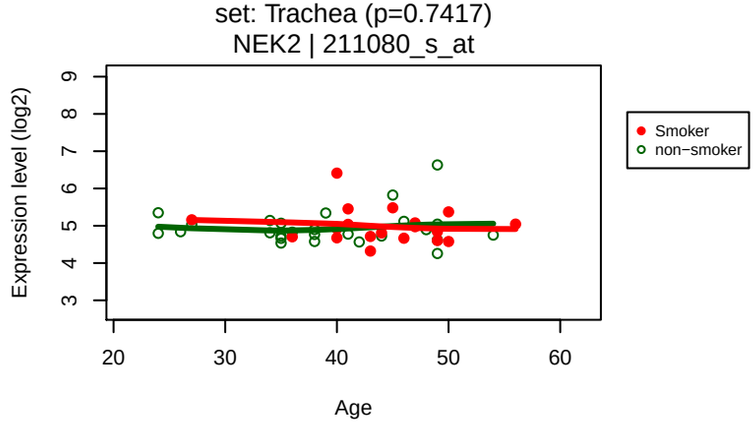

Gene expression ratio in Males
gene: NEK2 | 211080_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.069
p=0.6156
p=0.3761
p=0.1624
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK2 | 211080_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4868
p=0.2301
p=0.6323
p=0.8888
SmallAirway2
LargeAirway2
Trachea
Macrophage2

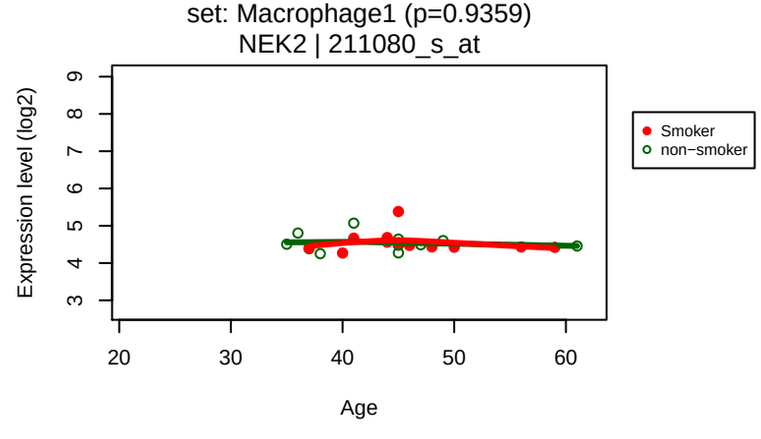

set: SmallAirway1 (p=0.0906)
NEK2 | 211080_s_at
Expression level (log2)
Age
Smoker
non−smoker

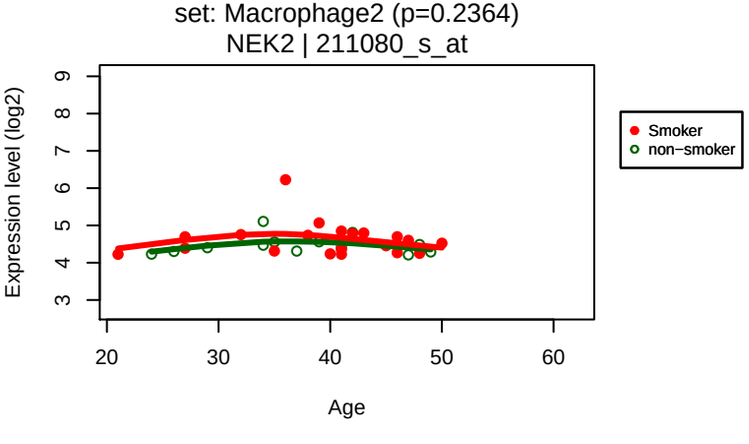

set: SmallAirway2 (p=0.0879)
NEK2 | 211080_s_at
Expression level (log2)
Age
Smoker
non−smoker

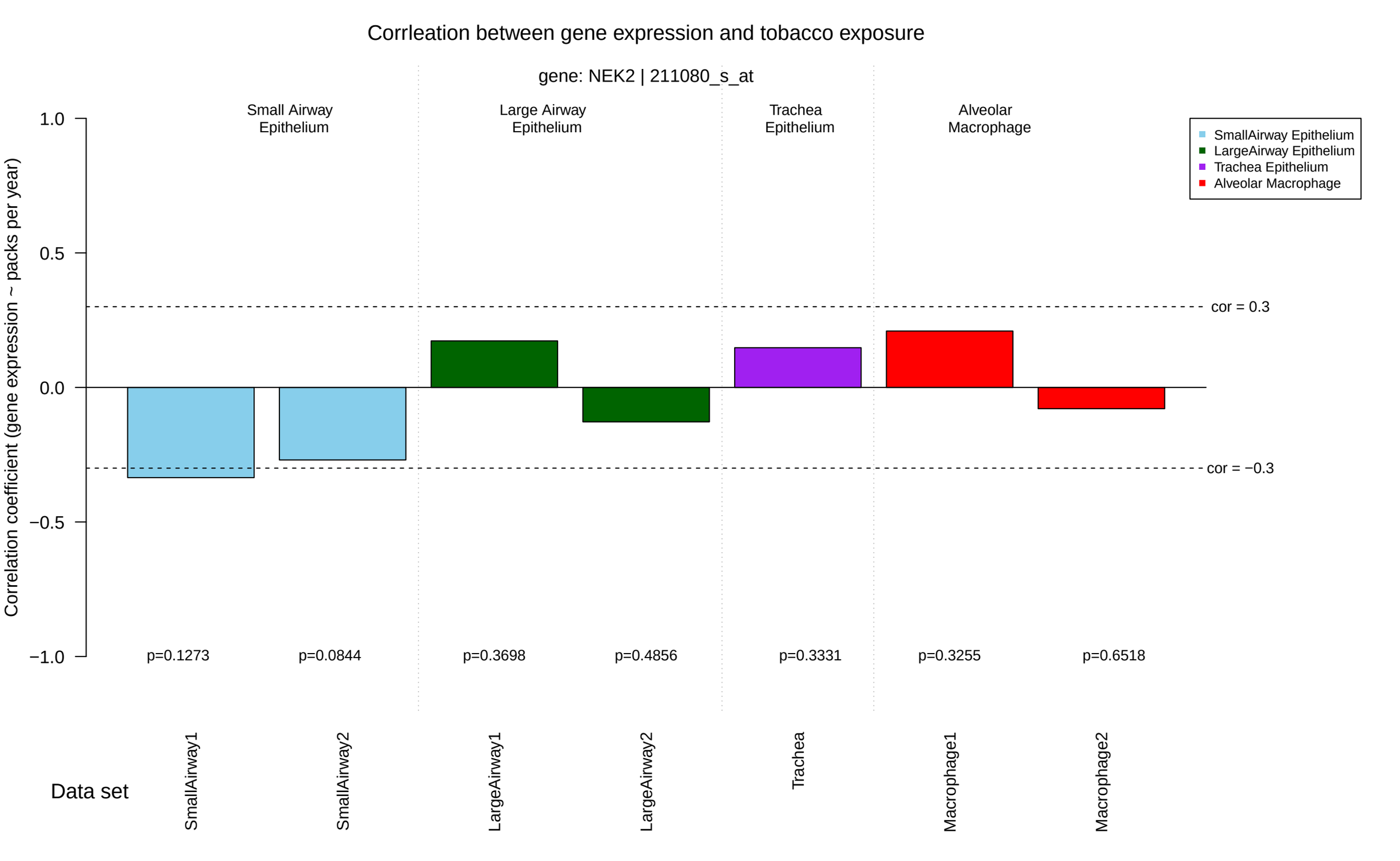

set: LargeAirway1 (p=0.4973)
NEK2 | 211080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5449)
NEK2 | 211080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7417)
NEK2 | 211080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9359)
NEK2 | 211080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2364)
NEK2 | 211080_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK2 | 211080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1273
p=0.0844
p=0.3698
p=0.4856
p=0.3331
p=0.3255
p=0.6518
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2